

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1677

Vc Porterdale Hydroelectric 138 kV
March 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1677. The Interconnection Customer proposes a 116.7 MW solar photovoltaic facility interconnecting at the Vc Porterdale Hydroelectric 138 kV substation in the Virginia area. The request was received on February 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 581 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$520,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
S Texa-Desert Outpost 138 kV	Brush Tavern-Rustburg 138 kV Line	102.8	503	7.9
Franklin-Elk Grove-New London Substation 138 kV line	Bear-Clover 345 kV	101.2	421	8.7
Nauvoo-Birchwood 345 kV	Fluvanna P.S-Louisa Generation Facility 345 kV	100.1	2,444	11.6
Bluff City-Lake Lure 138 kV	Louisa Generation Facility-Fluvanna P.S 345 kV line	100.0	847	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Texa-Desert Outpost 138 kV	Reconductor line	\$595,000	3% (\$20,000)
Franklin-Elk Grove-New London Substation 138 kV line	Reconductor line	\$748,000	3% (\$22,000)
Birchwood-Nauvoo 345 kV	Reconductor line	\$28,467,000	1% (\$391,000)
Bluff City-Lake Lure 138 kV	Reconductor line	\$4,995,000	2% (\$87,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$520,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.